

SIT215 Computational Intelligence

Assignment 2:

Intelligent System Design with Fuzzy Logic



Grading and weighting: (% total mark for unit): 20%

Submission Deadline: 8:00 pm AEST, 3 May 2026 (Week 8 Sunday)

Learning Objectives

This assignment requires you to demonstrate your achievement of unit learning outcomes ULO2 and ULO3 by:

- modelling real-world ambiguity using fuzzy representations
- designing meaningful fuzzy variables and membership functions
- constructing interpretable fuzzy rule bases, and
- reasoning about fuzzy system design and explaining its decision behaviour clearly and professionally

Assignment Overview

Transition from a deterministic route planning system for a truck delivery agent using search algorithms (Assignment 1), this assignment focuses on designing a fuzzy reasoning component that determines cargo handling or placement decisions in a delivery environment. The emphasis of this task is on:

- clear fuzzy system design
- interpretable rule-based reasoning
- justification of design choices

This assignment requires you to integrate concepts covered in Weeks 1 to 6, with particular emphasis on Weeks 5 and 6.

Note: Limited coding may be used to generate plots or diagrams and to compute numerical values for a worked example demonstration, while a full system implementation is not required for Assignment 2. The focus is on applying fuzzy logic principles to create a technically sound, context-appropriate design.

Submission

This is an individual assessment task. Please read this assignment tasksheet carefully. You **must** submit the main report, *[YourID]_assign2fuzzydesign_report.pdf* to Assignment 2 Dropbox on the Unit Site no later than the given deadline. Refer to sections [Your Tasks](#) and [Additional Guide for You to Complete this Assignment Effectively](#) for details of the criteria and required report structure.

Late Penalties

In accordance with Faculty assessment policies, late submissions to the submission folder will incur a penalty based on Deakin's Penalties for late submission and Due dates policies,

<https://www.deakin.edu.au/students/study-support/assessments-and-examinations/assessments>

Your Tasks

An Assignment 2 submission must present a clear and coherent fuzzy reasoning design for a delivery decision problem. All Tasks 1–3 are compulsory and will be evaluated together while the assignment is graded.

Task 1: Problem definition and fuzzy variables

- 1.1) Define a specific decision scenario related to delivery or cargo handling by selecting one of: stacking priority, temperature control, or urgency. Start your design with a mandatory variable: You must use *Cargo Fragility* (Input, Range 0 - 10).

Given: *Cargo Fragility* refers to the vulnerability of goods to damage, breakage, or degradation during shipping, handling, and transportation.

SAMPLE table structure:	Variable	Linguistic Labels	Numeric Range	Input/Output	Contextual Justification
	<i>Cargo Fragility</i>		from 0 to 10	Input	

- Your fuzzy design must include a total of three variables:
 - two fuzzy input variables, including the mandatory variable: *Cargo Fragility*, and one additional variable of your choice, and
 - one fuzzy output variable.
- At least three linguistic levels are defined for each variable.
- Each variable must define an appropriate numeric range associated with your defined linguistic labels.

1.2) In your report - section: *Problem Context and Fuzzy Variables*, specify your focusing delivery-related scenario and define all variables using a structured table. See the above **SAMPLE**. In the last column of the table, *Contextual Justification*, justify your choices of these labels and range against the real-world scenario. Add one new row to the table for each variable.

Task 2: Fuzzy system design with graphical representation

2.1) Design a fuzzy inference system that represents decision behaviour with the variables defined in Task 1. The fuzzy rules should represent trade-offs between multiple input conditions.

- Your design must include:
 - membership functions for all variables defined in Task 1
 - a fuzzy rule base using IF–THEN logic
 - a defuzzification and aggregation method

2.2) In your report - section: *Membership Function Design and Rules*,

- provide membership function plots for all variables. Each plot must include appropriate annotations/labels explaining:
 - where sets overlap
 - why such overlap is necessary for smooth reasoning
 - how the membership functions support the rule base
- present your logic using a **Rule Matrix**. Following the matrix, present your rule base with all meaningful combinations using IF–THEN logic. Then, name and justify your defuzzification method.

Task 3: Reasoning demonstration and evaluation

3.1) Demonstrate how your fuzzy system behaves using one example scenario chosen by you. Your example should show how input values are processed to produce the final decision for handling ambiguity.

3.2) In your report - section: *Worked Example*, present your picked worked example, using a structured table with rows covering the contexts along the five key stages. See the below **SAMPLE**. The aim of this table is to enable a fellow peer with a basic understanding of fuzzy systems to verify the reasoning and key technical process of your fuzzy system design. Therefore, you will have to decide on the level of detail to present.

SAMPLE table structure:	Key stage	Your example Value/logic	Technical process
	Crisp inputs	<i>Value/logic column: Provide specific numerical data, membership degrees, and rule-based logical steps of your selected worked example.</i>	<i>Technical process column: Explain how each value or logic outcome is inferred at each fuzzy reasoning stage, respectively</i>
	Fuzzification		
	Rule activation		
	Inference		
	Output		

SAMPLE of presenting numerical value | Crisp inputs | e.g. variable $a = 70$, variable $b = 50$.

To conclude this section, in 80 - 100 words and based on your given worked example, explain why the output decision is reasonable in your example, and how fuzzy reasoning helps handle uncertainty in this delivery context.

Marking

Your submission will be evaluated based on criteria that reflect the depth of clarity and quality of your fuzzy system design and reasoning. A numeric mark (out of a total mark of 20) will be determined by the submission quality against the rubric available on the Unit Site under Resources>Assessment.

Additional Guide for You to Complete this Assignment Effectively

Preparatory Learning Activities

To complete this assessment task effectively, you should apply computational intelligence and logical reasoning foundations introduced in earlier weeks of this unit, and showcase your mastery of Weeks 5 & 6 topics, such as fuzzy sets, linguistic variables, membership functions, rule base, inference and defuzzification. Engagement in lectures, practical exercises, and class discussions is highly recommended, which will benefit and support your design decisions.

Report Structure

Your report should be concise and structured, focusing on the key components of your fuzzy system design.

Sections	Guide for you		
Cover Page	Include your name, student ID, and the checklist (✓) below. The checklist helps you confirm that all required tasks are included in your submission.		
	Tasks	Completed (✓)	Notes (Optional)
	Task 1: Problem definition and fuzzy variables	☐	
	Task 2: Fuzzy system design with graphical representation	☐	
	Task 3: Reasoning demonstration and evaluation	☐	
Problem Context and Fuzzy Variables	Present the scenario and define fuzzy variables using the variable definition table from Task 1.		
Membership Function Design and Rules	Present the membership function design, annotated plots, rule matrix, and rule-based description, as described in Task 2.		
Worked Example	Present the reasoning demonstration described in Task 3, showing how input values lead to the final decision. Conclude the section with the design justification using the worked example.		
References	If applicable, all sources must be cited and referenced using a consistent academic referencing style.		

Expected Report Length

If completed correctly, the expected length is approximately 2 to 3 single-column A4 page(s), using a font size of 11 with single-line spacing, covering all essential sections as outlined. This assignment focuses on design clarity rather than lengthy explanations. Information should be presented using appropriate visual aids (e.g., tables, figures, diagrams, rule matrices, or flowcharts) to support your solution.

Getting Help and Support

Students are encouraged to engage in class (Lecture, Prac) discussions and exercises, and participate in the unit site discussion forum or Teams support channel, which can strengthen their understanding and contribute to peer learning. However, this is an individual assessment, and all development work and report writing must be your own. If using external references, ensure they are properly cited. Collusion and plagiarism will result in severe academic penalties. If you need clarification on how to complete this task, seek guidance from the teaching team.

Guidance Support

The teaching team is available to assist you with this task. If you need help understanding the material or requirements, seek support during class time or post your questions in the Discussion Forum on the Unit Site, where responses can also benefit your peers.

Report Writing Help

While the teaching team can offer guidance on report writing, ensure you review the instructions in this task sheet carefully. Additional support is available through the Writing Mentors program, which provides assistance with academic writing, including report preparation. For more information, visit <http://www.deakin.edu.au/students/studying/study-support/writing-mentors>.

Feedback

Students are encouraged to actively consolidate their design inspiration through peer discussions in class and to seek formative feedback from teaching staff on their work completed before the submission deadline, to ensure they are on track with this task. Formative feedback may be obtained during the weekly scheduled practical classes upon request. Talk to us in the classes, and we are happy to provide you with guidance! Furthermore, in 10 business days after the assignment submission due date, marking feedback will be provided via the unit site.

End of Assignment 2 Tasksheet.